

SFU

SIMON FRASER

UNIVERSITY

Theoretical Computing
Science Concentration**Fall 2024 GRADUA-
TION PLANNER****COMPUTING SCIENCE B.SC. MAJOR —**

Name: _____ Student ID: _____ Date: _____

Theoretical Computing Science covers algorithms, computability, complexity theory, and the mathematical foundations of computing.

Year 1

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|--|---|
| ☐ CMPT 120 Programming 1 | ☐ CMPT 125 Programming 2 |
| ☐ MACM 101 Discrete Math 1 | ☐ MATH 152 Calculus 2 |
| ☐ MATH 151 Calculus 1 | ☐ MATH 240 Algebra I: Linear Algebra |
| OR ☐ MATH 150 Calculus 1 with Review | OR ☐ MATH 232 Applied Linear Algebra |
| ☐ CMPT 105W CS Writing I (or in 2nd term) | ☐ WQB Breadth: _____ |
| ☐ WQB Breadth: _____ | ☐ General Elective: _____ |

Year 2

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|--|---|
| ☐ CMPT 225 Data Structures and Programming | ☐ CMPT 276 Software Engineering |
| ☐ CMPT 201 Introduction to Computer Science and Programming for Engineers / CS core | ☐ STAT 271 Probability and Statistics for CS |
| ☐ CMPT 295 Intro to Computer Systems | OR ☐ STAT 270 Introduction to Probability & Statistics |
| ☐ CMPT 210 Probability and Computing | ☐ WQB Breadth: _____ |
| OR ☐ MACM 201 Discrete Mathematics II | ☐ WQB Breadth: _____ |
| ☐ WQB Breadth: _____ | ☐ General Elective: _____ |

Year 3

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|--|---|
| ☐ CMPT 307 Data Structures and Algorithms | ☐ CMPT 300 Operating Systems |
| ☐ CMPT 308 Computability and Undecidability | ☐ MACM 300 Introduction to Complex Systems |
| ☐ MACM 316 Numerical Analysis | ☐ General Elective: _____ |
| ☐ CMPT 376W CS Writing II | ☐ General Elective: _____ |
| ☐ General Elective: _____ | ☐ General Elective: _____ |

Year 4

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|---|---|
| ☐ CMPT 405 Design and Analysis of Algorithms I | ☐ CMPT 404 Design and Analysis of Algorithms II* |
| ☐ CMPT 354 Database Systems | ☐ CMPT 406 Computational Complexity* |
| ☐ CMPT 379 Compilers | ☐ CMPT 407 Computational Geometry* |
| ☐ UD General Elective: _____ | ☐ UD General Elective: _____ |
| ☐ General Elective: _____ | ☐ General Elective: _____ |

* Or any other 400-level AI Elective: CMPT 410 Machine Learning, CMPT 411 Knowledge Representation, CMPT 412 Computational Vision, CMPT 413 Computational Linguistics, CMPT 414 Model-Based Computer Vision, CMPT 417 Intelligent Systems, CMPT 419 Special Topics in Artificial Intelligence, CMPT 420 Deep Learning.

Other recommended general electives: CMPT 320 Social Implications, STAT 350 Linear Models In Applied Statistics, STAT 440

WQB Breadth Requirements

- 6 units of Breadth Social (B-SOC)
- 6 units of Breadth Humanities (B-HUM)
- 3 units of Breadth Science (B-SCI)

Refer to http://www.sfu.ca/ugcr/for_students/wqb_requirements/breadth.html for courses that fulfill these requirements.

This Concentration Planning Form contains a recommended course plan for Computing Science major BSc students to obtain a concentration designation, along with course suggestions to optimize the knowledge and skills upon completion of this concentration, while distributing the difficulty of the course load per term. Other course plans may be possible. This form is not a substitute for the official degree regulations found at www.sfu.ca/students/calendar.html. If there is a question of interpretation or discrepancy, the University Calendar always takes precedence. For assistance or queries on possible substitutions, ask a FAS advisor to help. The student is ultimately responsible for ensuring that they have met their degree requirements.

CO-OPERATIVE EDUCATION Combines work experience with academic studies—all students are encouraged to apply once they have completed 30 units. Co-op does not count towards academic credits. Co-op is not mandatory; however, three work terms must be successfully completed in order to obtain an undergraduate degree with a co-op designation. For more information about Co-op, please see: <https://www.sfu.ca/coop/programs/cmpt/prospective.html>.

CMPT 415/416 SPECIAL RESEARCH PROJECTS are courses that may be used for upper division credit. See: <https://www.sfu.ca/computing/current-students/undergraduate-students/research.html>

FACULTY OF APPLIED SCIENCE RESIDENCY REQUIREMENTS At least two thirds of the total Upper Division (UD) units in the program must have been completed at Simon Fraser University. Please refer to the current SFU calendar for details.

CONTINUATION REQUIREMENTS Students who do not maintain at least a 2.40 CGPA, will be placed on probation by the School of Computing Science. Courses available to probationary students may be limited. Each term, these students must consult an advisor prior to enrollment and must achieve either a term 2.40 GPA or an improved CGPA. Students who fail to do so may be removed from the program.

ADVISING View drop-in advising times here <https://booking.cs.sfu.ca/adbooking/calendar.cgi> or email asadvise@sfu.ca. Please bring a copy of your advising transcript (download at go.sfu.ca) with you to the advising session.